

RF power meter specification comparison between R&S CMU200 and CMW500

Device	CMU200 (in wideband mode)	CMU200 (in frequency specific mode)	CMW500	FPC1500 w/ receiver option (FPC-K43)
Receiver architecture	Diode (6)	Receiver	Receiver. Diode only via external NRP-Z27 or NRP-Z28 series sensor and limited functionality (7)	Receiver
Frequency (or parameter) scan mode	No	No	Via "List" mode; up to 4000 steps (FW 3.0.12); each step can contain: • Freq. • Expected nom. power • Retrigger • Activate IQ amplitude acquisition (results via SCPI READ / FETCh, up to 4Msa • Parameter set (which contains trigger offset, step length, measurement length, filter type, RBW) Both vertical bars and table view Has auto list fill tool	In "Receiver" mode. Up to 10,000 steps of fixed internals (from 100Hz) that are calculated between the programmed min and max Both vertical bars and table view
Detector and basic statistics	• Peak (4) With bar chart (fixed range setting)	• Average • Min • Max All shown at the same time.	• "Average RMS" • Min • Max All shown at the same time.	• RMS • Average • Max • Quasi peak (maximum value weighted to CISPR 16-1-1) Two selected readings shown at the same time, both with their own bar chart (with various range settings. Has peak hold)
Sweep statistics	No	Statistical averages over multiple sweeps (1 to 1000 sweeps) • Average • Min • Max	Statistical averages over multiple sweeps (1 to 100.000 sweeps) • Min • Max • Standard Deviation	No
Units	dBm	dBm	dBm	dBm, dBmV, dBμV, V, W, dBμA Also linear scale
Measurement resolution shown for regular readings	0.1dBm or 0.01dBm resolution (5)	0.01dBm resolution	0.001dBm resolution	0.1dBm resolution
Frequency counter incoming signal	No	Yes (as long as signal is within the set RBW window)	No	No
Statistical likelihood distribution functions (graphical plots)	No	No	Cumulative Distribution Function (CCDF); Amplitude Probability Density (APD)	No
Power vs time	No	In "RF/Generator" then "Power". But limited to relatively short time windows (depend on RBW, for instance 19ms window for 300kHz RBW)	In "Spectrum" application, then "Zero Spam" All the way to 2000s	In spectrum analyzer mode, zero spam
Power vs frequency	n/a (wideband)	Has advanced spectrum analyzer	Has advanced spectrum analyzer	Has advanced spectrum analyzer
Freq range	100 kHz to 2700 MHz Has wideband measurement but also offers frequency allow for frequency response correction; can be set from 50 kHz (so even lower) to 2.7 GHz. (2)	10 MHz to 2700 MHz	70MHz to 6GHz (option KB036)	5 kHz to 3 GHz (8) (option FPC-B3)
Level range (in the shared 100 MHz to 3000 MHz range)	RF1: +6 dBm to +47 dBm (50 W), PEP: +53 dBm (200 W) RF2: -8 dBm to +33 dBm (2 W), PEP: +39 dBm (8 W) RF4: -33 dBm to 0 dBm	RF1: -40 dBm to +47 dBm (50 W), PEP: +53 dBm (200 W) RF2: -54 dBm to +33 dBm (2 W), PEP: +39 dBm (8 W) RF4: -80 dBm to 0 dBm	-84 dBm to +34 dBm (~2.5W) PEP: +42 dBm (~16W)	33 dBm (2 W); Peak power (<3 sec): 36 dBm (4 W) Manual attenuator Preamp
Uncertainty (in the shared 100 MHz to 3000 MHz range)	<0.5 dB	<0.5 dB	<0.5dB	Not published
Dynamic range (with fixed expected nominal power setting)	[About 40 dB]	[>74 dB ?] (spec in TDMA mode)	> 100 dB	Not published
Level resolution	0.1 dB (0.01 dB remote)	0.1 dB (0.01 dB remote)	0.01 dB	Not published
Level repeatability	Not published	Not published	< 0.01 dB	Not published
Resolution bandwidths (RBW)	n/a (RBW parameter *must* be set to 'WIDE')	10 Hz to 1 MHz in 1/2/3/5 steps	10 Hz to 10 MHz in 1/2/3/5 steps, plus 6.25 kHz	• 1 Hz to 3MHz in 1/3/10/30 steps (no in-between steps) • CISPR -6dB bandwidths of 200 Hz, 9 kHz, 120 kHz, 1 MHz, or AUTO
IF filter type for RBW	n/a (wideband)	With manual RBW setting: • Only Gaussian Also various standard-specific: • TDMA: root-raised cosine • CDMA: 1.4 MHz bandpass	With manual RBW setting: • Gaussian • Bandpass root-raised-cosine (RRC) Also various standard-specific (WCDMA, CDMA, TD-SCDMA)	Gaussian
Repetition	Measurement control: • Continuous • Single shot (button controlled)	Measurement control: • Continuous • Single shot (button controlled)		Measurement control: • Continuous scan • Single scan
Trigger	n/a	• Free • RF power • IF power • External (Note: you need an adapter cable from AUX3 DB15 connector!) Trigger parameters: • Trigger level -47 to 0dBm (IF, RF, not ext.) • Trigger slope	• Free run (3) • IF power • External • Dozens of standards-specific • Note: no RF power Trigger parameters: (9) • Trigger threshold (-50 to 0dBm) • Trigger slope • Trigger offset (1ns to 1s) • Trigger timeout (10ms to 3s) • Minimum trigger gap (10ns to 10ms) • Trigger mode (once, sweep, retrigger all, retrigger preselect)	(Not in receiver power measurement mode, only in SA Zero Spam mode,) • Free run • Video (~ IF power?) • External Trigger parameters: • Level in % (for video) • Trigger slope (for external) • Trigger delay 1μs to 100s (for external)
Aperture/timing related	[Fixed at unknown length?]	Fixed ('basic evaluation period' = 'sweep' is 4096 samples, duration unknown.	Measurement length @@@ (=aperture) Step length @@@ (must be longer than measurement)	5ms to 1000s
Expected input power settings	None, but there are three physical inputs with different level ranges (see above)	RF max level: -45dBm to +39dBm	• Expected nominal power: -55dBm to 55dBm (global or per list item) • User margin (= expected crest factor + reserve) (0-55dB) • Mixer level offset	None, but there is a manual attenuator.
Other			•	Upper and lower limits with beep and line in histogram

- (1) Evaluates a list of power steps at different levels and frequencies and performs statistical evaluation. ("Lists containing predefined levels and frequencies for GPRF generator/power meter test sequences")
- (2) Although diode-mode sensors measure RF power over a wide bandwidth, the correct RF signal frequency should be entered to achieve maximum accuracy and allow for frequency response correction.
- (3) In free run, several trigger settings are available, but they are irrelevant (as the manual confirms). Strange that they are not greyed out!
- (4) For actual wideband operation, you should either look at the reading in the Connection Control menu, or at the reading in the "Analyser Power" application, but in the latter, it is essential to set the RBW to "WIDE"; otherwise, it switches to a frequency selective (receiver architecture) reading and shows average rather than peak.
- (5) To see 0.01dBm resolution, go to the "Analyzer Power" application and make sure to RBW to "WIDE". If you do not do this, it actually performs a frequency-selective measurement using the receiving architecture!
- (6) Readings in the "Connection Control" menu, or in the "Analyser Power" application, and make sure to RBW to "WIDE". If you do not do this, it actually performs a frequency-selective measurement using the receiving architecture!
- (7) Available in the EPS application
- (8) Lower frequency values can be set (?), but seem to result in misreadings.
- (9) Can be selected in any trigger mode, but may not all be relevant in some modes like free run, and the threshold does not seem to match with the external input mode.